

# Momentum and Impulse — Complete Notes

Grade 12

Physical Sciences

CAPS

## What you'll learn

- Define momentum as a vector quantity and calculate it using  $p = m \cdot v$ .
- Explain impulse and apply the impulse–momentum theorem  $F_{\text{net}} \cdot \Delta t = \Delta p$ .
- State the principle of conservation of linear momentum and use it in isolated systems.
- Distinguish elastic from inelastic collisions using kinetic energy.
- Solve collision, rebound and explosion (recoil) problems with correct signs and units.
- Read momentum–time and force–time graphs.
- Avoid the common sign, unit and vector errors that cost marks in the examination.

## Momentum: the vector nature of motion

Every moving object has momentum. Momentum (symbol  $p$ ) is the product of an object's mass and its velocity. It measures how difficult an object is to stop: a heavy truck moving fast has a large momentum and is very hard to bring to rest.

Momentum is calculated with:

$$p = m \cdot v$$

- $p$  = momentum, measured in kilogram metre per second ( $\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ )
- $m$  = mass, measured in kilograms (kg)
- $v$  = velocity, measured in metres per second ( $\text{m} \cdot \text{s}^{-1}$ )

Because velocity is a vector, momentum is a vector. It has both a magnitude and a direction. This is the single most important idea in this topic, and it is where most learners lose marks. In every problem you must choose a positive direction at the start (for example, "to the right is positive") and then keep that sign convention for the whole calculation.

Momentum depends on velocity, not speed. Consider two identical cars, each moving at  $20 \text{ m} \cdot \text{s}^{-1}$ . One travels east and the other travels west. Their momenta have equal magnitude but opposite sign, because their velocities point in opposite directions.

### Sign convention

| Direction chosen as positive | Direction that becomes negative |
|------------------------------|---------------------------------|
| right, east, north, up       | left, west, south, down         |
| forwards                     | backwards                       |

## Change in momentum ( $\Delta p$ )

The change in momentum of an object is the difference between its final and initial momentum:

$$\Delta p = p_{\text{final}} - p_{\text{initial}} = m \cdot v_{\text{final}} - m \cdot v_{\text{initial}} = m \cdot \Delta v$$

Because it is a difference between two vectors, the change in momentum is also a vector. A ball that bounces straight back has a much larger change in momentum than a ball that simply stops, because its velocity

reverses direction, not just its magnitude.

## Impulse

An impulse is a force acting over a time interval. Impulse is defined as the product of the net force and the time for which it acts:

$$\text{Impulse} = F_{\text{net}} \cdot \Delta t$$

- Impulse is measured in newton seconds (N·s)
- $1 \text{ N} \cdot \text{s} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$ , so impulse and momentum have the same units

Newton's second law can be written in terms of momentum as  $F_{\text{net}} = \Delta p / \Delta t$ . Rearranging this gives the impulse–momentum theorem:

$$F_{\text{net}} \cdot \Delta t = \Delta p = m \cdot v_{\text{final}} - m \cdot v_{\text{initial}}$$

This theorem says that the impulse delivered to an object equals its change in momentum. It is the workhorse for any question that involves a force acting for a short time, such as a kick, a bat hitting a ball, or an airbag deploying.

### Safety applications: why crumple zones and airbags work

The change in momentum ( $\Delta p$ ) in a crash is fixed by the mass of the vehicle and its speed. Since  $F_{\text{net}} = \Delta p / \Delta t$ , if  $\Delta p$  is fixed, then a longer contact time  $\Delta t$  produces a smaller average force. Airbags, crumple zones, crash barriers, landing mats and even bending your knees when you land all work this way: they stretch out the time over which the object stops, which reduces the force on the people inside. This same reasoning explains why a cricketer's hands follow through when catching a fast ball.

## Conservation of linear momentum

Principle of conservation of linear momentum: the total linear momentum of an isolated system remains constant in both magnitude and direction.

An isolated system is a system on which the net external force is zero. During a collision or an explosion, the forces the objects exert on each other (internal forces) are equal and opposite by Newton's third law, and the interaction time is so short that external forces such as friction can be ignored. For such a system:

$$\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

This single equation solves collisions, rebounds, recoil and explosions. Write down all "before" momenta on the left, all "after" momenta on the right, and remember the signs.

### Elastic and inelastic collisions

| Type                | Momentum  | Kinetic energy | Notes                     |
|---------------------|-----------|----------------|---------------------------|
| Elastic             | conserved | conserved      | Ideal; e.g. gas molecules |
| Inelastic           | conserved | NOT conserved  | Most real collisions      |
| Perfectly inelastic | conserved | maximum loss   | Objects stick together    |

Kinetic energy is  $E_k = \frac{1}{2}mv^2$ . In an elastic collision both momentum and kinetic energy are conserved. In an inelastic collision momentum is conserved but some kinetic energy is converted to heat, sound and

deformation, so total kinetic energy decreases. When objects stick together after colliding, the collision is perfectly inelastic and they move on as a single mass ( $m_1 + m_2$ ).

### Worked example 1: Perfectly inelastic collision

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Two trolleys move toward each other on a frictionless track. Trolley A has a mass of 3 kg and moves to the right at  $4 \text{ m}\cdot\text{s}^{-1}$ . Trolley B has a mass of 5 kg and moves to the left at  $2 \text{ m}\cdot\text{s}^{-1}$ . They collide and stick together. Calculate their common velocity, and show that the collision is inelastic.

Step 1 — Choose a direction. Let to the right be positive.

Step 2 — Write down the momentum before the collision.

- $p_A = m \cdot v = (3)(+4) = +12 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
- $p_B = m \cdot v = (5)(-2) = -10 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
- $\text{total } p_{\text{before}} = 12 - 10 = +2 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$

Step 3 — Apply conservation of momentum. After sticking, the combined mass is  $3 + 5 = 8 \text{ kg}$  and they move with common velocity  $v$ .  $p_{\text{after}} = (8)(v) = 8v$

$$8v = 2, \text{ so } v = +0.25 \text{ m}\cdot\text{s}^{-1}$$

The common velocity is  $0.25 \text{ m}\cdot\text{s}^{-1}$  to the right.

Step 4 — Check the kinetic energy.

- $E_k \text{ before} = \frac{1}{2}(3)(4^2) + \frac{1}{2}(5)(2^2) = 24 + 10 = 34 \text{ J}$
- $E_k \text{ after} = \frac{1}{2}(8)(0.25^2) = 0.25 \text{ J}$

Kinetic energy is not conserved ( $34 \text{ J} \rightarrow 0.25 \text{ J}$ ), so the collision is inelastic.

### Worked example 2: Impulse and a rebound

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A soccer ball of mass  $0.45 \text{ kg}$  travels horizontally at  $12 \text{ m}\cdot\text{s}^{-1}$ . A player kicks it straight back along the same line, and it leaves the boot at  $20 \text{ m}\cdot\text{s}^{-1}$  in the opposite direction. The contact time is  $0.025 \text{ s}$ . Calculate the ball's change in momentum and the average net force on the ball.

Step 1 — Take the ball's original direction as positive.

Step 2 — Momentum before and after.

- $p_{\text{before}} = (0.45)(+12) = +5.4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
- $p_{\text{after}} = (0.45)(-20) = -9.0 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$

Step 3 — Change in momentum.  $\Delta p = p_{\text{after}} - p_{\text{before}} = -9.0 - 5.4 = -14.4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$

Step 4 — Average net force using  $F_{\text{net}} = \Delta p / \Delta t$ .  $F_{\text{net}} = (-14.4) / (0.025) = -576 \text{ N}$

The magnitude of the force is  $576 \text{ N}$ , and the negative sign means it acts opposite to the ball's original direction — exactly what we expect, because the boot reverses the ball.

### Worked example 3: Explosion (recoil)

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A cannon of mass  $1200 \text{ kg}$  fires a shell of mass  $15 \text{ kg}$  horizontally at  $400 \text{ m}\cdot\text{s}^{-1}$ . Calculate the recoil velocity of the cannon.

Step 1 — Take the direction the shell travels as positive.

Step 2 — The cannon and shell are at rest before firing, so total momentum before = 0.

Step 3 — Apply conservation of momentum.  $0 = (15)(+400) + (1200)(v)$   $0 = 6000 + 1200v$   $1200v = -6000$   $v = -5 \text{ m}\cdot\text{s}^{-1}$

The cannon recoils at  $5 \text{ m}\cdot\text{s}^{-1}$  in the direction opposite to the shell.

## Graphs

- The gradient of a momentum–time graph equals the net force (because  $F_{\text{net}} = \Delta p / \Delta t$ ).
- The area under a force–time graph equals the impulse, which equals  $\Delta p$ .
- On a momentum–time graph, a horizontal line means constant momentum and therefore zero net force.

## Common mistakes

- Treating momentum as a scalar and ignoring the direction sign for objects moving in opposite directions.
- Mixing units: mass must be in kg (convert grams), and speed must be in  $\text{m}\cdot\text{s}^{-1}$  (convert  $\text{km}\cdot\text{h}^{-1}$  by dividing by 3.6).
- Writing a momentum answer with the unit N or J. Momentum is measured in  $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$ .
- Assuming kinetic energy is conserved in every collision. Only elastic collisions conserve it.
- Using the same sign for the velocity before and after a rebound, which hides the reversal.
- Forgetting that in a "stick together" collision the masses add.
- Confusing impulse ( $F\cdot\Delta t$ , in N·s) with force (in N) or with work and energy (in J).

## Quick summary / cheat sheet

|                                                                          |                                         |                                              |
|--------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|
| $p = m \cdot v$                                                          | momentum                                | $(\text{kg}\cdot\text{m}\cdot\text{s}^{-1})$ |
| $\Delta p = m \cdot \Delta v = m(v_{\text{final}} - v_{\text{initial}})$ | change in momentum                      | $(\text{kg}\cdot\text{m}\cdot\text{s}^{-1})$ |
| $F_{\text{net}} \cdot \Delta t = \Delta p$                               | impulse-momentum theorem                | $(\text{N}\cdot\text{s})$                    |
| $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$                     | conservation of momentum                | (isolated system)                            |
| $E_k = \frac{1}{2}mv^2$                                                  | kinetic energy                          | $(\text{J})$                                 |
| Elastic:                                                                 | momentum and $E_k$ both conserved       |                                              |
| Inelastic:                                                               | momentum conserved, $E_k$ NOT conserved |                                              |
| Stick together:                                                          | inelastic, use one mass ( $m_1 + m_2$ ) |                                              |

Conversions and tips:  $1 \text{ km}\cdot\text{h}^{-1} = 1/3.6 \text{ m}\cdot\text{s}^{-1}$ . To reduce the force on an object during a fixed change in momentum, increase the contact time  $\Delta t$ .

## Practice questions

1. A 1500 kg car moves east at  $20 \text{ m}\cdot\text{s}^{-1}$ . Calculate its momentum, giving the magnitude and direction.
2. A 0.5 kg ball is thrown vertically downwards at  $8 \text{ m}\cdot\text{s}^{-1}$  and bounces straight back up at  $6 \text{ m}\cdot\text{s}^{-1}$ . Take downwards as positive and calculate the change in the ball's momentum.
3. Two trolleys on a frictionless track: A (2 kg) moves right at  $5 \text{ m}\cdot\text{s}^{-1}$  and B (3 kg) moves left at  $2 \text{ m}\cdot\text{s}^{-1}$ . They collide and stick together. Find their common velocity and state whether the collision is elastic or inelastic.
4. A net force of 240 N acts on a 4 kg object that is initially at rest for 0.5 s. Calculate the impulse and the final velocity of the object.

5. A 0.06 kg arrow leaves a 3 kg bow at  $80 \text{ m}\cdot\text{s}^{-1}$ . Calculate the recoil velocity of the bow.
6. Explain, using physics, why a gymnast bends her knees when she lands after a jump.

### Answers

1.  $p = m \cdot v = (1500)(20) = 3 \times 10^4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$  towards the east.
2. Downwards is positive.  $p_{\text{before}} = (0.5)(8) = 4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ . After the bounce the velocity is upwards, so it is negative:  $p_{\text{after}} = (0.5)(-6) = -3 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ .  $\Delta p = -3 - 4 = -7 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ , that is  $7 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$  upwards.
3. Let right be positive.  $p_{\text{before}} = (2)(5) + (3)(-2) = 10 - 6 = 4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ .  $p_{\text{after}} = (2 + 3)v = 5v$ , so  $5v = 4$  and  $v = 0.8 \text{ m}\cdot\text{s}^{-1}$  to the right.  $E_k \text{ before} = \frac{1}{2}(2)(25) + \frac{1}{2}(3)(4) = 25 + 6 = 31 \text{ J}$ ;  $E_k \text{ after} = \frac{1}{2}(5)(0.8^2) = 1.6 \text{ J}$ . Kinetic energy is not conserved, so the collision is inelastic.
4. Impulse =  $F_{\text{net}} \cdot \Delta t = (240)(0.5) = 120 \text{ N}\cdot\text{s}$ . This equals  $\Delta p$ , so  $\Delta p = 120 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ . Since the object started from rest,  $\Delta p = m \cdot v$ , giving  $v = 120/4 = 30 \text{ m}\cdot\text{s}^{-1}$ .
5. Total momentum before = 0.  $0 = (0.06)(80) + (3)(v) = 4.8 + 3v$ , so  $v = -1.6 \text{ m}\cdot\text{s}^{-1}$ . The bow recoils at  $1.6 \text{ m}\cdot\text{s}^{-1}$  in the opposite direction to the arrow.
6. Bending the knees increases the time  $\Delta t$  over which the body comes to rest. The change in momentum  $\Delta p$  is fixed by the landing speed and mass, and because  $F_{\text{net}} = \Delta p / \Delta t$ , a larger  $\Delta t$  produces a smaller average net force on the gymnast's legs. This reduces the risk of injury.

Official EduRank Team notes. Always cross-check with your textbook and teacher.